

Jyoti Yadav

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EDUCATION

Vellore Institute of Technology, *B.Tech in Computer Science and Engineering* | Chennai

2028

Kendriya Vidyalaya, *CBSE Class XII*

94.2% ,2024

SUMMARY

Third-year CSE student with hands-on experience in software engineering, data analytics, and AI/ML projects. Skilled in Python, SQL, backend development, and machine learning, with experience building data-driven and scalable applications. Seeking a Summer Internship to apply technical and analytical skills to real-world business problems.

EXPERIENCE

Laparoscopic Smoke Removal, *Research Work* | Under Geetha S.

2026 - Present

- Conducting research on a real-time physics-guided latent diffusion framework for laparoscopic smoke removal using temporal feature fusion.
- Integrating foundation vision models with diffusion-based restoration to preserve surgical structures under challenging smoke conditions.
- Exploring temporal feature fusion across consecutive frames for consistent and artifact-free smoke removal.
- Investigating physics-guided constraints to improve reconstruction fidelity and real-time restoration performance.

CodeChef, *Competitive Programmer* | *Project Member*

2025 – Present

- Active competitive programmer solving algorithmic problems and participating in coding contests.
- Contributed to an Agentic Honeypot for intelligent threat detection and attack analysis.

SKILLS

Languages & Technologies

C, C++, Java, JavaScript, TypeScript, Python, SQL, Node.js, Express, FastAPI, React, Next.js, PostgreSQL, MySQL, Oracle, MongoDB, Pandas, NumPy, Docker, Jenkins, REST APIs, Streamlit

Artificial Intelligence & Machine Learning

Scikit-learn, TensorFlow, OpenCV, YOLOv8, CNNs, LSTMs, Computer Vision, NLP, RAG, FAISS

Projects

BizERP – *Full Stack Business ERP Application*

- Built a full-stack ERP system for managing employees, customers, inventory, sales, and procurement.
- Developed secure REST APIs using ASP.NET Core, Entity Framework Core, JWT authentication, and role-based access control.
- Designed SQL Server data models and transactional workflows for consistent inventory, sales, and procurement operations.
- Built responsive React dashboards with Tailwind CSS for real-time business insights, reporting, and inventory tracking.

LinkX – *Serverless URL Shortener*

- Built a URL shortener supporting custom aliases, redirects, and click tracking.
- Developed REST APIs using Hono and TypeScript with modular service design.
- Integrated PostgreSQL with Prisma ORM and implemented Zod-based input validation.
- Built a React and Tailwind CSS frontend for interacting with backend services.

DocMind – *ML-Powered Document Q&A*

- Built an AI-powered document Q&A system using BERT, FAISS, Sentence Transformers, and Ollama for private RAG-based retrieval.
- Implemented semantic chunking, vector search, and cross-encoder reranking to improve retrieval accuracy.
- Developed confidence-based answer generation with source citations to reduce hallucinations.
- Integrated OCR and document classification for processing diverse PDF and scanned documents.

Achievements

- **Branch Academic Rank:** Ranked **4th** in 1st Year and **10th** in 2nd Year
- **LeetCode:** 600+ problems solved | Rating: **1706** (Max: 1749) | Rating: **1890** global rank in Weekly Contest 517
- **CodeChef:** Starters 225 (Rated) | Global Rank: **409**
- **CBSE Class X:** Scored **97%**, placing among the **Top 1.5%** of students.